

COUNTEREXAMPLES TO A HIGHER-INDEX DIRICHLET EIGENVALUE-RATIO CONJECTURE

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ABSTRACT. Let $\lambda_k(\Omega)$ denote the k th Dirichlet Laplacian eigenvalue of a bounded planar domain Ω , with eigenvalues counted with multiplicity. We disprove a conjectured higher-index extension of the Payne–Pólya–Weinberger inequality that was recorded as an open problem by Ashbaugh. For every integer $m \geq 3$, we construct a bounded planar domain Ω_m with C^∞ boundary such that

$$\frac{\lambda_{2m}(\Omega_m)}{\lambda_m(\Omega_m)} > \frac{13}{5} > \frac{j_{1,1}^2}{j_{0,1}^2},$$

where $j_{\nu,1}$ is the first positive zero of J_ν , the Bessel function of the first kind of order ν . For $m = 3$, the annulus $\{x \in \mathbb{R}^2 : 1/10 < |x| < 1\}$ already provides such a counterexample.

1. INTRODUCTION

Throughout the paper, a *domain* is a bounded connected open subset of Euclidean space. We write $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. For any bounded open set $G \subset \mathbb{R}^N$, not necessarily connected, let $\lambda_j(G)$ denote the j th eigenvalue of the Dirichlet Laplacian, with eigenvalues counted with multiplicity. In particular, if G is a domain, then $0 < \lambda_1(G) < \lambda_2(G) \leq \lambda_3(G) \leq \dots$. If J_ν denotes the Bessel function of the first kind of order ν and $j_{\nu,q}$ its q th positive zero, define

$$\gamma_N := \frac{j_{N/2,1}^2}{j_{N/2-1,1}^2}.$$

Payne, Pólya, and Weinberger initiated the systematic study of universal ratios of Dirichlet eigenvalues. In the planar case they conjectured that the ratio λ_2/λ_1 is maximized by the disk, and their 1956 paper established classical non-sharp estimates that motivated the problem [15]. The corresponding sharp inequality in dimension N is

$$\frac{\lambda_2(\Omega)}{\lambda_1(\Omega)} \leq \gamma_N. \tag{1}$$

Ashbaugh and Benguria proved (1) in all dimensions [3]; the constant is sharp and is attained by balls.

The higher-index question considered here grew out of the same circle of ideas. Ashbaugh and Benguria proved, in particular, that $\lambda_4(\Omega)/\lambda_2(\Omega) < \gamma_N$ for connected domains, as a special case of a more general estimate involving the number of nodal domains of an eigenfunction [4]. They also obtained the iterated bound $\lambda_{2^r}(\Omega)/\lambda_1(\Omega) \leq \gamma_N^r$ [5]. Ashbaugh subsequently included the corresponding one-step inequality in his list of open problems [2, Problem 3, p. 15]. The same problem was recorded in the 2007 and 2009 Oberwolfach reports. Those reports noted that the cases $m = 1, 2$ were known and that the cases $m \geq 3$ remained open [7, p. 1021]; see also [6, pp. 400–401]. The conjecture can be stated as follows.

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Conjecture 1.1. *For every integer $m \geq 1$, every $N \geq 2$, and every domain $\Omega \subset \mathbb{R}^N$,*

$$\frac{\lambda_{2m}(\Omega)}{\lambda_m(\Omega)} \leq \gamma_N. \quad (2)$$

Our main result disproves the conjecture for every $m \geq 3$, already in the plane.

Theorem 1.2. *For every integer $m \geq 3$, there exists a planar domain Ω_m with C^∞ boundary such that*

$$\frac{\lambda_{2m}(\Omega_m)}{\lambda_m(\Omega_m)} > \frac{13}{5}.$$

For $m = 3$, one may take the circular annulus $A := \{x \in \mathbb{R}^2 : 1/10 < |x| < 1\}$.

Lemma 2.2 gives $\gamma_2 < 13/5$. Consequently, Theorem 1.2 gives strict counterexamples to (2). It also disproves, for $N = 2$ and $k = 2$, the natural more general estimate

$$\frac{\lambda_{km}(\Omega)}{\lambda_m(\Omega)} \leq C_{N,k}, \quad C_{N,k} := \sup_D \frac{\lambda_k(D)}{\lambda_1(D)},$$

where the supremum is taken over domains $D \subset \mathbb{R}^N$. This broader formulation was also discussed in Ashbaugh's open-problem survey [2, p. 15]. Recent non-sharp estimates for ratios of arbitrary Dirichlet eigenvalues on convex domains may be found in [9]; our examples are nonconvex and do not address possible restrictions of Conjecture 1.1 to narrower geometric classes.

The proof is explicit. Separation of variables reduces the annular spectrum to one-dimensional weighted problems. Polynomial trial functions and a localized Hardy estimate identify the first seven annular eigenvalues and show that $\lambda_6(A)/\lambda_3(A) > 13/5$. All numerical comparisons are certified by exact arithmetic and rational bounds, including Euler–Rayleigh bounds for Bessel zeros, so no floating-point computation enters the proof. To treat every $m \geq 4$, we adjoin $m - 3$ rectangles whose first three eigenvalues interlace with the relevant annular eigenvalues in the required way. Exact spectral counting on the disconnected union gives the same ratio, and standard spectral convergence under shrinking connections then produces smooth connected examples.

Section 2 develops the Bessel-zero certificates. Section 3 analyzes the annulus and proves the case $m = 3$. Section 4 amplifies the construction to every $m \geq 4$. Section 5 explains why the usual nodal-domain argument stops short of the conjectured index. Exact trial-function integrals and the remaining elementary rational estimates are collected in the appendix.

2. CERTIFIED BOUNDS FOR BESSEL ZEROS

We use the following elementary form of the Euler–Rayleigh inequalities; see, for example, [12, Lemma 3.2]. The product formula for J_ν and the Rayleigh sums of its zeros are standard; see [14, §10.21(iii),(xiii)].

Lemma 2.1. *Let $x_1 > x_2 > \cdots > 0$ and suppose that $S_p := \sum_{q \geq 1} x_q^p < \infty$. Then, for every integer $p \geq 1$,*

$$S_p^{-1/p} < \frac{1}{x_1} < \frac{S_p}{S_{p+1}}.$$

Proof. The first inequality follows from $S_p > x_1^p$. Since $x_q < x_1$ for $q \geq 2$, one also has $S_{p+1} < x_1 S_p$, which gives the second inequality. $\square$

For $\nu \geq 0$ and $p \geq 1$, define the *Rayleigh sum* $\sigma_p(\nu) := \sum_{q=1}^{\infty} j_{\nu,q}^{-2p}$. Applying Lemma 2.1 to $x_q = j_{\nu,q}^{-2}$ gives

$$\sigma_p(\nu)^{-1/p} < j_{\nu,1}^2 < \frac{\sigma_p(\nu)}{\sigma_{p+1}(\nu)}. \quad (3)$$

The Rayleigh sums satisfy the classical convolution recurrence of Kishore [13]; see also the Riccati-equation derivation and generalizations of Gupta and Muldoon [10]:

$$\sigma_1(\nu) = \frac{1}{4(\nu+1)}, \quad \sigma_p(\nu) = \frac{1}{\nu+p} \sum_{r=1}^{p-1} \sigma_r(\nu) \sigma_{p-r}(\nu) \quad (p \geq 2). \quad (4)$$

The values needed below are listed in Table 1; each follows by repeated substitution in (4). These computations may also be verified by the accompanying exact-arithmetic script [11].

TABLE 1. Rayleigh sums used in the proof.

ν	p	$\sigma_p(\nu)$
0	2	1/32
1	3	1/3072
1	5	13/8847360
1	6	11/110100480
3	5	1/110100480

Lemma 2.2. *The first Bessel zeros satisfy*

$$j_{1,1}^2 > 14, \quad j_{3,1}^2 > \frac{1014}{25}, \quad \gamma_2 = \frac{j_{1,1}^2}{j_{0,1}^2} < \frac{13}{5}.$$

Proof. The lower bound in (3) gives $j_{1,1}^2 > 3072^{1/3} > 14$, since $3072 > 14^3$. It also gives

$$j_{3,1}^2 > 110100480^{1/5} > \frac{1014}{25},$$

because $110100480 \cdot 25^5 - 1014^5 = 3212367382176 > 0$.

For the ratio, the lower and upper bounds in (3) give

$$j_{0,1}^2 > \sqrt{32}, \quad j_{1,1}^2 < \frac{\sigma_5(1)}{\sigma_6(1)} = \frac{1456}{99}.$$

Hence

$$\gamma_2 = \frac{j_{1,1}^2}{j_{0,1}^2} < \frac{1456}{99\sqrt{32}} < \frac{13}{5},$$

where the last inequality follows after squaring from $13^2 \cdot 99^2 \cdot 32 - 5^2 \cdot 1456^2 = 5408 > 0$. $\square$

3. AN ANNULAR COUNTEREXAMPLE

Set $a := 1/10$ and $A := \{x \in \mathbb{R}^2 : a < |x| < 1\}$. For $n \in \mathbb{N}_0$ and $f \in H_0^1(a, 1) \setminus \{0\}$, define

$$\mathcal{Q}_n[f] := \frac{\int_a^1 \left(r|f'(r)|^2 + \frac{n^2}{r}|f(r)|^2 \right) dr}{\int_a^1 r|f(r)|^2 dr}. \quad (5)$$

This is precisely the Rayleigh quotient of the Dirichlet Laplacian on functions of the form $u(r, \theta) = f(r)e^{in\theta}$; see, for example, [1, (2.6) and Appendix A]. Let $\mu_{n,q}$ be the q th variational eigenvalue associated with (5). Separation of variables then shows that the Dirichlet spectrum of A is the multiset consisting of $\mu_{0,q}$ with multiplicity one and $\mu_{n,q}$ with multiplicity two for every $n \geq 1$.

3.1. Angular monotonicity and comparison with the disk.

Lemma 3.1. *For every $n \geq 0$ and $q \geq 1$,*

$$\mu_{n+1,q} \geq \mu_{n,q} + 2n + 1. \quad (6)$$

Proof. Since $a < r < 1$, one has $r^{-1} \geq r$. Hence, for every $f \in H_0^1(a, 1) \setminus \{0\}$,

$$\mathcal{Q}_{n+1}[f] - \mathcal{Q}_n[f] = (2n + 1) \frac{\int_a^1 |f|^2 r^{-1} dr}{\int_a^1 |f|^2 r dr} \geq 2n + 1.$$

Taking the min-max over q -dimensional subspaces of $H_0^1(a, 1)$ gives (6). $\square$

Lemma 3.2. *For every $n \geq 0$, one has $\mu_{n,1} \geq j_{n,1}^2$.*

Proof. Extend $f \in H_0^1(a, 1)$ by zero from $(a, 1)$ to $(0, 1)$. This embeds the form domain of the annular n th angular sector into the corresponding form domain for the unit disk. The min-max principle gives $\mu_{n,1} \geq j_{n,1}^2$. $\square$

3.2. Separation from the second radial mode. The only step in identifying the low-lying spectrum that does not follow immediately from monotonicity is to show that the first $n = 3$ mode lies below the second radial mode.

Proposition 3.3. *One has $\mu_{3,1} < \mu_{0,2}$.*

Proof. Consider the trial function $f_3(r) := (r - a)(1 - r)(1 - 7r + 4r^2)$. The exact integral evaluations recorded in Appendix A.1 give

$$\mathcal{Q}_3[f_3] = \frac{7000(10^6 \log 10 + 12031677)}{2451519837}.$$

Using $\log 10 < 2303/1000$ from Lemma A.1, we obtain

$$\mu_{3,1} \leq \mathcal{Q}_3[f_3] < \frac{100342739000}{2451519837} < 41. \quad (7)$$

We next estimate the second radial eigenvalue from below. Put $v(r) := r^{1/2}f(r)$. Integration by parts, using $v(a) = v(1) = 0$, gives

$$\int_a^1 r |f'|^2 dr = \int_a^1 \left(|v'|^2 - \frac{|v|^2}{4r^2} \right) dr, \quad \int_a^1 r |f|^2 dr = \int_a^1 |v|^2 dr.$$

Choose $c := 391/1000$. From $v(r) = \int_a^r v'(s) ds$ on (a, c) and $v(r) = -\int_r^1 v'(s) ds$ on $(c, 1)$, Cauchy-Schwarz and Fubini's theorem give

$$\begin{aligned} \int_a^c \frac{|v|^2}{4r^2} dr &\leq C_L \int_a^c |v'|^2 dr, & C_L &:= \frac{1}{4} \left(\log \frac{c}{a} + \frac{a}{c} - 1 \right), \\ \int_c^1 \frac{|v|^2}{4r^2} dr &\leq C_R \int_c^1 |v'|^2 dr, & C_R &:= \frac{1}{4} \left(\frac{1}{c} + \log c - 1 \right). \end{aligned}$$

The logarithmic estimates in Lemma A.1 imply $C_L < 39/250$ and $C_R < 39/250$. Therefore

$$\int_a^1 \left(|v'|^2 - \frac{|v|^2}{4r^2} \right) dr > \frac{211}{250} \int_a^1 |v'|^2 dr. \quad (8)$$

The min–max principle and the Dirichlet spectrum of the interval $(a, 1)$ now give

$$\mu_{0,2} \geq \frac{211}{250} \left(\frac{2\pi}{1-a} \right)^2 = \frac{211}{250} \left(\frac{20\pi}{9} \right)^2.$$

Using $\pi > 157/50$ from Lemma A.1, we conclude that

$$\mu_{0,2} > \frac{10401878}{253125} > 41. \quad (9)$$

The bounds (7) and (9) prove the claim. $\square$

Corollary 3.4. *The first seven eigenvalues of A are*

$$\begin{aligned} \lambda_1(A) &= \mu_{0,1}, & \lambda_2(A) &= \lambda_3(A) = \mu_{1,1}, \\ \lambda_4(A) &= \lambda_5(A) = \mu_{2,1}, & \lambda_6(A) &= \lambda_7(A) = \mu_{3,1}. \end{aligned} \quad (10)$$

Proof. Lemma 3.1 gives $\mu_{0,1} < \mu_{1,1} < \mu_{2,1} < \mu_{3,1} < \mu_{4,1} < \dots$. For every $n \geq 0$ and $q \geq 2$, angular monotonicity gives $\mu_{n,q} \geq \mu_{0,q} \geq \mu_{0,2} > \mu_{3,1}$, where the last inequality is Proposition 3.3. Taking angular multiplicities into account yields (10). $\square$

3.3. Eigenvalue-ratio estimate for the annulus. Define $\alpha := \mu_{1,1} = \lambda_3(A)$ and $\beta := \mu_{3,1} = \lambda_6(A)$.

Proposition 3.5. *The annulus A satisfies*

$$\frac{\lambda_6(A)}{\lambda_3(A)} = \frac{\beta}{\alpha} > \frac{13}{5}.$$

Proof. Lemmas 3.2 and 2.2 give $\beta = \mu_{3,1} \geq j_{3,1}^2 > 1014/25$. For the denominator, take $f_1(r) := (r-a)(1-r)(1-r/2)$. Appendix A.1 gives

$$\mathcal{Q}_1[f_1] = \frac{56(800000 \log 10 + 4831011)}{23993577}.$$

The estimate $\log 10 < 2303/1000$ yields

$$\alpha \leq \mathcal{Q}_1[f_1] < \frac{373711016}{23993577} < \frac{78}{5}.$$

It follows that

$$\frac{\beta}{\alpha} > \frac{1014/25}{78/5} = \frac{13}{5}. \quad \square$$

Lemma 2.2 gives $\gamma_2 < 13/5$, so Proposition 3.5 proves Theorem 1.2 for $m = 3$.

4. COUNTEREXAMPLES FOR ALL HIGHER INDICES

We first record the standard approximation that turns a finite disjoint union into a smooth connected domain without changing any fixed finite part of its Dirichlet spectrum in the limit.

Lemma 4.1. *Let $G^{(0)} \subset \mathbb{R}^2$ be a finite disjoint union of bounded Lipschitz domains. There exists a sequence of bounded connected domains G_ℓ with C^∞ boundary such that, for every fixed $j \geq 1$,*

$$\lambda_j(G_\ell) \longrightarrow \lambda_j(G^{(0)}) \quad (\ell \rightarrow \infty).$$

Proof. Write $G^{(0)} = D_1 \dot{\cup} \cdots \dot{\cup} D_s$. Choose connected smooth domains $D_i^{(r)} \subset\subset D_i$ increasing to D_i , and set

$$G^{(r)} := \dot{\bigcup}_{i=1}^s D_i^{(r)}.$$

Standard domain-approximation results give $\lambda_j(G^{(r)}) \rightarrow \lambda_j(G^{(0)})$ ($r \rightarrow \infty$) for every fixed j ; see [8, Proposition 7.1, Corollary 4.7, and Remark 4.3].

Fix r . After rigid motions of the components, which do not change their spectra, arrange them in a chain with pairwise disjoint interiors, so that consecutive closures meet at finitely many points and all other closures are disjoint. Let E be the resulting finite set of contact points. By smoothing the contacts inside small neighborhoods of E , we obtain bounded connected C^∞ domains $G_\varepsilon^{(r)}$, all contained in a fixed bounded set, such that

$$G^{(r)} \subset G_\varepsilon^{(r)} \subset G^{(r)} \cup N_{C\varepsilon}(E), \quad N_\rho(E) := \{x : \text{dist}(x, E) < \rho\}, \quad (11)$$

where C is independent of ε .

We verify the three hypotheses of [8, Theorem 7.5] along an arbitrary sequence $\varepsilon_n \downarrow 0$. First, if $K \subset\subset G^{(r)}$, then $K \setminus G_{\varepsilon_n}^{(r)} = \emptyset$ by the first inclusion in (11), and therefore $\text{cap}(K \setminus G_{\varepsilon_n}^{(r)}) = 0$. Here cap denotes the standard $(1, 2)$ -capacity in $\mathbb{R}^2$,

$$\text{cap}(F) := \inf \left\{ \|v\|_{H^1(\mathbb{R}^2)}^2 : v \in H^1(\mathbb{R}^2), v \geq 1 \text{ a.e. in a neighborhood of } F \right\},$$

as in [8, p. 605]. For the second hypothesis, the single open set $U := \mathbb{R}^2 \setminus \overline{G^{(r)}}$ covers $\mathbb{R}^2 \setminus \overline{G^{(r)}}$. Since E is finite, the second inclusion in (11) gives

$$|U \cap G_{\varepsilon_n}^{(r)}| \leq |N_{C\varepsilon_n}(E)| = O(\varepsilon_n^2) \rightarrow 0.$$

Therefore, by [8, Proposition 7.6(1)], $\lambda_1(U \cap G_{\varepsilon_n}^{(r)}) \rightarrow \infty$. Finally, the boundary accumulation set appearing in [8, (7.2)], namely

$$\Gamma := \bigcap_{n=1}^{\infty} \overline{\bigcup_{k \geq n} (G_{\varepsilon_k}^{(r)} \cap \partial G^{(r)})},$$

is contained in E by (11). Since $\Gamma \subset E$ and E is finite, we have $\text{cap}(\Gamma) = 0$. Consequently, $H_0^1(G^{(r)} \cup \Gamma) = H_0^1(G^{(r)})$, which is the third hypothesis of [8, Theorem 7.5]. Therefore, the convergence conclusion of [8, Theorem 7.5] applies to $G_\varepsilon^{(r)}$ as $\varepsilon \downarrow 0$. Since all these domains lie in a fixed bounded set, [8, Corollary 4.7 and Remark 4.3] yields $\lambda_j(G_\varepsilon^{(r)}) \rightarrow \lambda_j(G^{(r)})$ ($\varepsilon \downarrow 0$) for every fixed j , counting multiplicities. A diagonal choice of $r \rightarrow \infty$ and $\varepsilon \downarrow 0$ completes the proof. $\square$

We now choose an auxiliary rectangle with exactly one eigenvalue below α and exactly two below β . Let

$$P := \left(0, \frac{\pi}{2}\right) \times \left(0, \frac{\pi}{\sqrt{10}}\right).$$

Its separated Dirichlet eigenvalues are $4p^2 + 10q^2$, where $p, q \in \mathbb{N}$, and hence

$$\lambda_1(P) = 14, \quad \lambda_2(P) = 26, \quad \lambda_3(P) = 44.$$

The estimates above give the transparent chain

$$14 < \alpha < \frac{78}{5} < 26 < \frac{1014}{25} < \beta < 41 < 44. \quad (12)$$

In particular, $\lambda_1(P) < \alpha < \lambda_2(P) < \beta < \lambda_3(P)$.

Proposition 4.2. *For every integer $m \geq 4$, there exists a planar domain Ω_m with C^∞ boundary such that*

$$\frac{\lambda_{2m}(\Omega_m)}{\lambda_m(\Omega_m)} > \frac{13}{5}.$$

Proof. Let $P_1, \dots, P_{m-3}$ be disjoint copies of P and set

$$G_m^{(0)} := A \dot{\cup} P_1 \dot{\cup} \dots \dot{\cup} P_{m-3}.$$

The spectrum of a finite disjoint union is the multiset union of the spectra of its components. By (10) and (12), there are exactly $1 + (m - 3) = m - 2$ eigenvalues of $G_m^{(0)}$ strictly below α , while α has multiplicity two. Hence $\lambda_m(G_m^{(0)}) = \alpha$. Likewise, there are exactly $5 + 2(m - 3) = 2m - 1$ eigenvalues strictly below β , while β has multiplicity two. Thus

$$\lambda_m(G_m^{(0)}) = \alpha, \quad \lambda_{2m}(G_m^{(0)}) = \beta, \quad \frac{\lambda_{2m}(G_m^{(0)})}{\lambda_m(G_m^{(0)})} > \frac{13}{5}. \quad (13)$$

Apply Lemma 4.1 to $G_m^{(0)}$. The m th and $2m$ th eigenvalues of the resulting smooth connected domains converge to α and β , respectively. Since the last inequality in (13) is strict, the same strict inequality holds for all sufficiently large members of the approximating sequence. Choose one such domain and denote it by Ω_m . $\square$

Propositions 3.5 and 4.2 complete the proof of Theorem 1.2.

5. RELATION TO NODAL-DOMAIN ESTIMATES

The mechanism behind the known $m = 2$ case also explains why the conjecture was plausible. Suppose that an eigenfunction u for $\lambda_m(\Omega)$ has $\ell(u)$ nodal domains. On every nodal domain D_i , the restriction of u is a first Dirichlet eigenfunction, so $\lambda_1(D_i) = \lambda_m(\Omega)$. Applying the definition of $C_{N,k}$ to each D_i , taking the disjoint union, and using Dirichlet domain monotonicity gives

$$\lambda_{k\ell(u)}(\Omega) \leq C_{N,k} \lambda_m(\Omega). \quad (14)$$

This argument, in a sharper Bessel-zero form, is already present in Ashbaugh and Benguria's work on higher eigenvalue ratios [4].

For $m = 2$, every eigenfunction associated with $\lambda_2(\Omega)$ on a connected domain has exactly two nodal domains, and (14) with $k = 2$ gives the required index 4. The annulus makes the obstruction at $m = 3$ transparent. Its λ_3 -eigenspace is $f(r) \text{span}\{\cos \theta, \sin \theta\}$, where f is the positive first radial eigenfunction in the $n = 1$ sector. Every nonzero eigenfunction in this space has exactly two nodal domains, not three. Thus the nodal argument controls $\lambda_4(A)$ at the scale $\gamma_2 \lambda_3(A)$, whereas Conjecture 1.1 asks for control of $\lambda_6(A)$. Proposition 3.5 shows that this missing index shift is not merely a limitation of the method: the claimed inequality itself fails.

APPENDIX A. EXACT-ARITHMETIC VERIFICATION

For reproducibility, the exact computations in this appendix, as well as the Rayleigh sums in Table 1, can be independently verified using the accompanying Python/SymPy script [11]. The repository also contains the output of a successful run. The script verifies the trial-function integrals and the rational certificates used below using symbolic and exact rational arithmetic only; no floating-point computation is involved.

A.1. Exact trial-function integrals. For $f \in H_0^1(a, 1)$, set

$$\begin{aligned} I_0(f) &:= \int_a^1 r f(r)^2 dr, & I_1(f) &:= \int_a^1 r f'(r)^2 dr, \\ I_{-1}(f) &:= \int_a^1 \frac{f(r)^2}{r} dr. \end{aligned}$$

For the two trial functions used above, direct expansion and integration give

$$\begin{aligned} I_0(f_1) &= \frac{23993577}{4480000000}, & I_1(f_1) &= \frac{2401569}{40000000}, \\ I_{-1}(f_1) &= \frac{\log 10}{100} + \frac{27873}{80000000}, \\ I_0(f_3) &= \frac{22063678533}{700000000000}, & I_1(f_3) &= \frac{187046577}{350000000}, \\ I_{-1}(f_3) &= \frac{\log 10}{100} + \frac{42655833}{700000000}. \end{aligned}$$

Substitution into $\mathcal{Q}_n[f] = (I_1(f) + n^2 I_{-1}(f))/I_0(f)$ gives the quotient formulas used in Propositions 3.3 and 3.5.

A.2. Elementary bounds for constants.

Lemma A.1. *The following inequalities hold:*

$$\log 10 < \frac{2303}{1000}, \quad \pi > \frac{157}{50}, \quad \log \frac{391}{100} < \frac{273}{200}, \quad \log \frac{391}{1000} < -\frac{187}{200}.$$

Proof. The first inequality follows from the Taylor series of the exponential:

$$\exp\left(\frac{2303}{1000}\right) > \sum_{q=0}^9 \frac{(2303/1000)^q}{q!} > 10.$$

For the second, Machin's formula and the alternating series for the arctangent give

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239} > 4 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} \right) - \frac{1}{239} = \frac{70369}{89625} > \frac{157}{200}.$$

For $x > 0$, put $t := (x - 1)/(x + 1)$. Then $\log x = 2 \sum_{q=0}^{\infty} t^{2q+1}/(2q+1)$. For $x = 391/100$, one has $t = 291/491$. Bounding the positive tail after the t^5 term by a geometric series yields

$$\log \frac{391}{100} \leq 2 \left(t + \frac{t^3}{3} + \frac{t^5}{5} + \frac{t^7}{7(1-t^2)} \right) < \frac{273}{200}.$$

For $x = 391/1000$, one has $t = -609/1391$. All terms in the series are negative, and hence

$$\log \frac{391}{1000} < 2 \left(t + \frac{t^3}{3} + \frac{t^5}{5} \right) < -\frac{187}{200}.$$

The final comparisons are exact rational inequalities. $\square$

We finish by checking the constants used in (8). Since $100/391 < 32/125$ and $1000/391 < 1279/500$, the preceding lemma gives

$$\begin{aligned} C_L &< \frac{1}{4} \left(\frac{273}{200} + \frac{32}{125} - 1 \right) = \frac{621}{4000} < \frac{39}{250}, \\ C_R &< \frac{1}{4} \left(\frac{1279}{500} - \frac{187}{200} - 1 \right) = \frac{623}{4000} < \frac{39}{250}. \end{aligned}$$

STATEMENT ON AI USAGE

We acknowledge the use of AI during the development of this work. The authors first identified the annulus

$$\{x \in \mathbb{R}^2 : 1/10 < |x| < 1\}$$

as a candidate counterexample for $m = 3$. ChatGPT assisted in finding and carrying out the explicit estimates needed to turn this candidate into a rigorous counterexample, including suitable trial functions and exact numerical certificates. The extension of the construction from $m = 3$ to every $m \geq 4$, using auxiliary rectangles with suitably interlacing Dirichlet eigenvalues and the resulting spectral-counting argument in Section 4, was obtained with ChatGPT.

The accompanying Python/SymPy code for verifying the exact-arithmetic computations [11] was also generated by ChatGPT. All AI-assisted mathematical arguments, computations, and code were subsequently carefully checked by the authors. The authors take full responsibility for all mathematical claims and computational results in this paper.

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